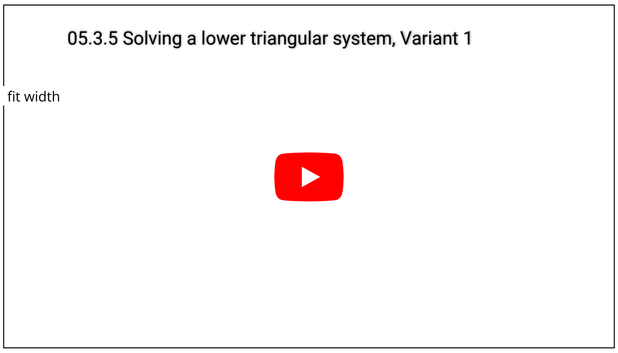
5.3.5 Solving with a triangular matrix

We are left to discuss how to solve $Lz = y$ and $Ux = z$.

5.3.5.1 Algorithmic Variant 1



Consider $Lz = y$ where L is unit lower triangular. Partition

$$L \rightarrow \left(\begin{array}{c c} 1 & 0 \\ l_{21} & L_{22} \end{array}\right), \quad z \rightarrow \left(\begin{array}{c} \zeta_1 \\ z_2 \end{array}\right) \quad \text{and} \quad y \rightarrow \left(\begin{array}{c} \psi_1 \\ y_2 \end{array}\right).$$

Then

$$\underbrace{\left(\begin{array}{c c} 1 & 0 \\ l_{21} & L_{22} \end{array}\right)}_{L} \underbrace{\left(\begin{array}{c} \zeta_1 \\ z_2 \end{array}\right)}_{z} = \underbrace{\left(\begin{array}{c} \psi_1 \\ y_2 \end{array}\right)}_{y}.$$

Multiplying out the left-hand side yields

$$\left(\begin{array}{c} \zeta_1 \\ \zeta_1 l_{21} + L_{22} z_2 \end{array}\right) = \left(\begin{array}{c} \psi_1 \\ y_2 \end{array}\right)$$

and the equalities

$$\begin{array}{rcl} \zeta_1 & = & \psi_1 \\ \zeta_1 l_{21} + L_{22} z_2 & = & y_2, \end{array}$$

which can be rearranged as

$$\begin{array}{rcl} \zeta_1 & = & \psi_1 \\ L_{22} z_2 & = & y_2 - \zeta_1 l_{21}. \end{array}$$

We conclude that in the current iteration

- ψ_1 needs not be updated.
- $y_2 := y_2 - \psi_1 l_{21}$

So that in future iterations $L_{22} z_2 = y_2$ (updated!) will be solved, updating z_2.

These insights justify the algorithm in Figure 5.3.5.1, which overwrites y with the solution to $Lz = y$.

Solve $Lz = y$, overwriting y with z (Variant 1)

$$L \rightarrow \left(\begin{array}{c|c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array}\right), y \rightarrow \left(\begin{array}{c} y_T \\ \hline y_B \end{array}\right)$$

L_{TL} is 0×0 and y_T has 0 elements

while $n(L_{TL}) < n(L)$

$$\left(\begin{array}{c|c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array}\right) \rightarrow \left(\begin{array}{c|c c} L_{00} & l_{01} & L_{02} \\ \hline l_{10}^T & \lambda_{11} & l_{12}^T \\ L_{20} & l_{21} & L_{22} \end{array}\right), \left(\begin{array}{c} y_T \\ \hline y_B \end{array}\right) \rightarrow \left(\begin{array}{c} y_0 \\ \hline \psi_1 \\ y_2 \end{array}\right)$$

$$y_2 := y_2 - \psi_1 l_{21}$$

$$\left(\begin{array}{c|c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array}\right) \leftarrow \left(\begin{array}{c c|c} L_{00} & l_{01} & L_{02} \\ l_{10}^T & \lambda_{11} & l_{12}^T \\ \hline L_{20} & l_{21} & L_{22} \end{array}\right), \left(\begin{array}{c} y_T \\ \hline y_B \end{array}\right) \leftarrow \left(\begin{array}{c} y_0 \\ \psi_1 \\ \hline y_2 \end{array}\right)$$

endwhile

Figure 5.3.5.1. Lower triangular solve (with unit lower triangular matrix), Variant 1

Homework 5.3.5.1. Derive a similar algorithm for solving $Ux = z$. Update the below skeleton algorithm with the result. (Don't forget to put in the lines that indicate how you "partition and repartition" through the matrix.)

Solve $Ux = z$, overwriting z with x (Variant 1)

$$U \rightarrow \left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array} \right), z \rightarrow \left(\begin{array}{c} z_T \\ \hline z_B \end{array} \right)$$

U_{BR} is 0×0 and z_B has 0 elements

while $n(U_{BR}) < n(U)$

$$\left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array} \right) \rightarrow \left(\begin{array}{ccc} U_{00} & u_{01} & U_{02} \\ u_{10}^T & \upsilon_{11} & u_{12}^T \\ U_{20} & u_{21} & U_{22} \end{array} \right), \left(\begin{array}{c} z_T \\ \hline z_B \end{array} \right) \rightarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ z_2 \end{array} \right)$$

$$\left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array} \right) \leftarrow \left(\begin{array}{ccc} U_{00} & u_{01} & U_{02} \\ u_{10}^T & \upsilon_{11} & u_{12}^T \\ U_{20} & u_{21} & U_{22} \end{array} \right), \left(\begin{array}{c} z_T \\ \hline z_B \end{array} \right) \leftarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ z_2 \end{array} \right)$$

endwhile

Hint: Partition

$$\left(\begin{array}{cc} U_{00} & u_{01} \\ 0 & \upsilon_{11} \end{array} \right) \left(\begin{array}{c} x_0 \\ \chi_1 \end{array} \right) = \left(\begin{array}{c} z_0 \\ \zeta_1 \end{array} \right).$$

▾ Solution

Multiplying this out yields

$$\left(\begin{array}{c} U_{00}x_0 + u_{01}\chi_1 \\ \upsilon_{11}\chi_1 \end{array} \right) = \left(\begin{array}{c} z_0 \\ \zeta_1 \end{array} \right).$$

So, $\chi_1 = \zeta_1/\upsilon_{11}$ after which x_0 can be computed by solving $U_{00}x_0 = z_0 - \chi_1 u_{01}$. The resulting algorithm is then given by

Solve $Ux = z$, overwriting z with x (Variant 1)

$$U \rightarrow \left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array} \right), z \rightarrow \left(\begin{array}{c} z_T \\ \hline z_B \end{array} \right)$$

U_{BR} is 0×0 and z_B has 0 elements

while $n(U_{BR}) < n(U)$

$$\left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array} \right) \rightarrow \left(\begin{array}{cc|c} U_{00} & u_{01} & U_{02} \\ u_{10}^T & \upsilon_{11} & u_{12}^T \\ \hline U_{20} & u_{21} & U_{22} \end{array} \right), \left(\begin{array}{c} z_T \\ \hline z_B \end{array} \right) \rightarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ \hline z_2 \end{array} \right)$$

$\zeta_1 := \zeta_1/\upsilon_{11}$

$z_0 := z_0 - \zeta_1 u_{01}$

$$\left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c|cc} U_{00} & u_{01} & U_{02} \\ \hline u_{10}^T & \upsilon_{11} & u_{12}^T \\ U_{20} & u_{21} & U_{22} \end{array} \right), \left(\begin{array}{c} z_T \\ \hline z_B \end{array} \right) \leftarrow \left(\begin{array}{c} z_0 \\ \hline \zeta_1 \\ z_2 \end{array} \right)$$

endwhile

5.3.5.2 Algorithmic Variant 2 ¶

05.3.5 Solving a lower triangular system, Variant 2

fit width

An alternative algorithm can be derived as follows: Partition

$$L \rightarrow \left(\begin{array}{cc} L_{00} & 0 \\ l_{10}^T & 1 \end{array} \right), \quad z \rightarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \end{array} \right) \quad \text{and} \quad y \rightarrow \left(\begin{array}{c} y_0 \\ \psi_1 \end{array} \right).$$

Then

$$\underbrace{\left(\begin{array}{cc} L_{00} & 0 \\ l_{10}^T & 1 \end{array} \right)}_{L} \underbrace{\left(\begin{array}{c} z_0 \\ \zeta_1 \end{array} \right)}_{z} = \underbrace{\left(\begin{array}{c} y_0 \\ \psi_1 \end{array} \right)}_{y}.$$

Multiplying out the left-hand side yields

$$\left(\begin{array}{c} L_{00}z_0 \\ l_{10}^T z_0 + \zeta_1 \end{array} \right) = \left(\begin{array}{c} y_0 \\ \psi_1 \end{array} \right)$$

and the equalities

$$\begin{array}{rcl} L_{00}z_0 & = & y_0 \\ l_{10}^T z_0 + \zeta_1 & = & \psi_1. \end{array}$$

The idea now is as follows: Assume that the elements of z_0 were computed in previous iterations in the algorithm in Figure 5.3.5.2, overwriting y_0. Then in the current iteration we must compute $\zeta_1 := \psi_1 - l_{10}^T z_0$, overwriting ψ_1.

Solve $Lz = y$, overwriting y with z (Variant 2)

$$L \rightarrow \left(\begin{array}{c|c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array}\right), y \rightarrow \left(\begin{array}{c} y_T \\ \hline y_B \end{array}\right)$$

L_{TL} is 0×0 and y_T has 0 elements

while $n(L_{TL}) < n(L)$

$$\left(\begin{array}{c|c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array}\right) \rightarrow \left(\begin{array}{c|cc} L_{00} & l_{01} & L_{02} \\ \hline l_{10}^T & \lambda_{11} & l_{12}^T \\ L_{20} & l_{21} & L_{22} \end{array}\right), \left(\begin{array}{c} y_T \\ \hline y_B \end{array}\right) \rightarrow \left(\begin{array}{c} y_0 \\ \hline \psi_1 \\ y_2 \end{array}\right)$$

$\psi_1 := \psi_1 - l_{10}^T y_0$

$$\left(\begin{array}{c|c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array}\right) \leftarrow \left(\begin{array}{cc|c} L_{00} & l_{01} & L_{02} \\ l_{10}^T & \lambda_{11} & l_{12}^T \\ \hline L_{20} & l_{21} & L_{22} \end{array}\right), \left(\begin{array}{c} y_T \\ \hline y_B \end{array}\right) \leftarrow \left(\begin{array}{c} y_0 \\ \psi_1 \\ \hline y_2 \end{array}\right)$$

endwhile

Figure 5.3.5.2. Lower triangular solve (with unit lower triangular matrix), Variant 2

Homework 5.3.5.2. Derive a similar algorithm for solving $Ux = z$. Update the below skeleton algorithm with the result. (Don't forget to put in the lines that indicate how you "partition and repartition" through the matrix.)

Solve $Ux = z$, overwriting z with x (Variant 2)

$$U \rightarrow \left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array}\right), z \rightarrow \left(\begin{array}{c} z_T \\ \hline z_B \end{array}\right)$$

U_{BR} is 0×0 and z_B has 0 elements

while $n(U_{BR}) < n(U)$

$$\left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array}\right) \rightarrow \left(\begin{array}{ccc} U_{00} & u_{01} & U_{02} \\ u_{10}^T & \upsilon_{11} & u_{12}^T \\ U_{20} & u_{21} & U_{22} \end{array}\right), \left(\begin{array}{c} z_T \\ \hline z_B \end{array}\right) \rightarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ z_2 \end{array}\right)$$

$$\left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array}\right) \leftarrow \left(\begin{array}{ccc} U_{00} & u_{01} & U_{02} \\ u_{10}^T & \upsilon_{11} & u_{12}^T \\ U_{20} & u_{21} & U_{22} \end{array}\right), \left(\begin{array}{c} z_T \\ \hline z_B \end{array}\right) \leftarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ z_2 \end{array}\right)$$

endwhile

Hint: Partition

$$U \rightarrow \left(\begin{array}{cc} \upsilon_{11} & u_{12}^T \\ 0 & U_{22} \end{array}\right).$$

▾ Solution

Partition

$$\left(\begin{array}{cc} \upsilon_{11} & u_{12}^T \\ 0 & U_{22} \end{array}\right)\left(\begin{array}{c} \chi_1 \\ x_2 \end{array}\right) = \left(\begin{array}{c} \zeta_1 \\ z_2 \end{array}\right).$$

Multiplying this out yields

$$\left(\begin{array}{c} \upsilon_{11}\chi_1 + u_{12}^T x_2 \\ U_{22} x_2 \end{array}\right) = \left(\begin{array}{c} \zeta_1 \\ z_2 \end{array}\right).$$

So, if we assume that x_2 has already been computed and has overwritten z_2, then χ_1 can be computed as

$$\chi_1 = (\zeta_1 - u_{12}^T x_2)/\upsilon_{11}$$

which can then overwrite ζ_1. The resulting algorithm is given by

Solve $Ux = z$, overwriting z with x (Variant 2)

$$U \rightarrow \left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array}\right), z \rightarrow \left(\begin{array}{c} z_T \\ \hline z_B \end{array}\right)$$

U_{BR} is 0×0 and z_B has 0 elements

while $n(U_{BR}) < n(U)$

$$\left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array}\right) \rightarrow \left(\begin{array}{cc|c} U_{00} & u_{01} & U_{02} \\ u_{10}^T & \upsilon_{11} & u_{12}^T \\ \hline U_{20} & u_{21} & U_{22} \end{array}\right), \left(\begin{array}{c} z_T \\ \hline z_B \end{array}\right) \rightarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ \hline z_2 \end{array}\right)$$

$\zeta_1 := \zeta_1 - u_{12}^T z_2$

$\zeta_1 := \zeta_1 / \upsilon_{11}$

$$\left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array}\right) \leftarrow \left(\begin{array}{c|cc} U_{00} & u_{01} & U_{02} \\ \hline u_{10}^T & \upsilon_{11} & u_{12}^T \\ U_{20} & u_{21} & U_{22} \end{array}\right), \left(\begin{array}{c} z_T \\ \hline z_B \end{array}\right) \leftarrow \left(\begin{array}{c} z_0 \\ \hline \zeta_1 \\ z_2 \end{array}\right)$$

endwhile

Homework 5.3.5.3. Let L be an $m \times m$ unit lower triangular matrix. If a multiply and add each require one flop, what is the approximate cost of solving $Lx = y$?

▾ Solution

Let us analyze Variant 1.

Let L_{00} be $k \times k$ in a typical iteration. Then y_2 is of size $m - k - 1$ and $y_2 := y_2 - \psi_1 l_{21}$ requires $2(m - k - 1)$ flops. Summing this over all iterations requires

$$\sum_{k=0}^{m-1}[2(m-k-1)] \text{ flops.}$$

The change of variables $j = m - k - 1$ yields

$$\sum_{k=0}^{m-1}[2(m-k-1)] = 2\sum_{j=0}^{m-1} j \approx m^2.$$

Thus, the cost is approximately m^2 flops.

5.3.5.3 Discussion ¶

Computation tends to be more efficient when matrices are accessed by column, since in scientific computing applications tend to store matrices by columns (in column-major order). This dates back to the days when Fortran ruled supreme. Accessing memory consecutively improves performance, so computing with columns tends to be more efficient than computing with rows.

Variant 1 for each of the algorithms casts computation in terms of columns of the matrix that is involved;

$$y_2 := y_2 - \psi_1 l_{21}$$

and

$$z_0 := z_0 - \zeta_1 u_{01}.$$

These are called ***axpy*** operations:

$$y := \alpha x + y.$$

"*a*lpha times *x* *p*lus *y*." In contrast, Variant 2 casts computation in terms of rows of the matrix that is involved:

$$\psi_1 := \psi_1 - l_{10}^T y_0$$

and

$$\zeta_1 := \zeta_1 - u_{12}^T z_2$$

perform dot products.